

Nara Ávila

Master's Student

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Education

M.Sc. in Nuclear Physics [\[link\]](#)

2025 – Present

Physics Institute (IF), University of São Paulo (USP)

My research focuses on the properties of quark–gluon plasma (QGP) created in relativistic heavy-ion collisions at the Large Hadron Collider (LHC) and the Relativistic Heavy Ion Collider (RHIC). Working within a research group specializing in heavy-ion physics and relativistic hydrodynamics, I use Bayesian model-to-data inference to constrain parameters describing the QGP. In particular, I investigate how correlations in experimental uncertainties affect parameter estimation and conclusions about the medium's properties, while exploring quantum computing methods to infer these correlations. My project combines computational modeling, machine-learning methods—including Gaussian-process emulation—and Markov chain Monte Carlo sampling to connect theoretical predictions with experimental measurements from heavy-ion collisions.

B.Sc. in Computer Engineering [\[link\]](#)

2006 – 2011

São Carlos School of Engineering (EESC), University of São Paulo (USP)

The interdisciplinary curriculum encompassed four major areas of computer engineering:

Computer Science & Software Engineering: Algorithms and data structures, compilers, object-oriented and distributed programming, software engineering, UML, design patterns, MVC, software testing, relational databases and SQL, linear optimization, and performance evaluation.

Computer Architecture: Digital logic, microprocessors, pipelined RISC/CISC architectures, parallel computing, SIMD/MIMD systems, and shared- and distributed-memory organization. Hardware design included VHDL, Verilog, programmable logic, and ASIC synthesis and layout using LeonardoSpectrum and Mentor tools.

Physics & Electronics: Classical and quantum mechanics, electromagnetism, circuit analysis, semiconductor devices, analog and digital electronics, CMOS microelectronics, digital signal processing, Fourier and Z transforms, digital filters, and microprocessor-based control systems.

Communications & Networks: Communication theory, modulation, antennas, optical communications, TCP/IP, routing, TCP/UDP, wireless networks, and quality of service. Practical work included Java video streaming with RTSP/RTP, IEEE 802.11 analysis with Wireshark, and Android networking with Bluetooth and Wi-Fi.

Academic Positions

Research Collaborator [\[link\]](#)

2023 – 2025

Department of Materials Physics and Mechanics (DFMT), Institute of Physics (IF), University of São Paulo (USP)

40 hours/week

Project: Quantum Computing Applied to Molecular Modeling

Research Context: Investigation of quantum computing methods for first-principles molecular modeling in materials and energy research. The work focused on computing ground-state energies of molecules relevant to the energy transition, including H₂ and CO₂, as a first step toward the use of quantum simulations in the discovery and development of new materials and sustainable energy technologies.

Quantum Simulation of Molecular Systems: Developed first-principles quantum simulations for molecular ground-state energy estimation. Electronic-structure problems were formulated within the Born–Oppenheimer approximation and second quantization, with fermionic Hamiltonians mapped onto qubit operators using the Jordan–Wigner transformation. Ground-state energies were estimated with VQE using UCCSD ansätze and benchmarked against Hartree–Fock and Full Configuration Interaction (FCI) calculations. The study examined basis-set selection, active-space approximations, shot counts, and qubit-reduction strategies, and explored restricted Boltzmann machines (RBMs) as alternative variational representations of molecular quantum states.

Quantum Computing Methods & Platforms: Implemented and evaluated gate-based quantum algorithms across Amazon Braket, IBM Qiskit, Google Cirq, CERN Qibo, and PennyLane, considering superconducting, trapped-ion, photonic, and neutral-atom architectures. Hardware-oriented studies examined execution time, noise, and decoherence, together with NISQ error mitigation using zero-noise extrapolation (ZNE) and Clifford data regression (CDR).

Teaching Assistant [\[link\]](#) *Fall 2009*

Institute of Mathematics and Computer Sciences (ICMC), University of São Paulo (USP)

8 hours/week

- SME0300 Numerical Methods I

Teaching Assistant [\[link\]](#) *Spring 2009*

Institute of Mathematics and Computer Sciences (ICMC), University of São Paulo (USP)

8 hours/week

- SME0601 Numerical Methods II

Academic Awards

• Best Presentation in Environmental Management System (SHS0436) *Fall 2009*

- Recognized for an impactful final course presentation entitled “Harmonic and Disharmonic Systems,” due to its combination of visual simplicity and strong conceptual content. Using images alone, the presentation explored in detail the diversity, complexity, and beauty of harmonic systems in nature—including coral ecosystems and harmonious ecological relationships such as mutualism, commensalism, and epiphytism—in contrast with human systems that disrupt environmental balance, illustrated through examples of waste accumulation, pollution, environmental degradation, and pollution-related wildlife mortality, alongside other disharmonic relationships also found in nature, such as parasitism and disease. Received the maximum final grade (10.0/10.0).

• 9th place – ACM-ICPC / SBC Brazilian Programming Contest, First Phase [\[link\]](#) *Sep 2007*

Campinas

- Competed as a member of team “Greve de Acertos?” in the first phase of the Brazilian Programming Contest (Maratona de Programação), the Brazilian regional competition of the ACM International Collegiate Programming Contest (ICPC).

• 6th place in the IPhO / SBF Brazilian Physics Olympiad [\[link\]](#) *Oct 2003*

- National physics olympiad organized by the Brazilian Physical Society (SBF) and part of Brazil’s selection process for the International Physics Olympiad (IPhO).

• 18th place in the OBM / XI OMEG [\[link\]](#) *Sep 2002*

- Mathematics olympiad competition associated with the Brazilian Mathematical Olympiad (OBM), Brazil’s national mathematics olympiad and part of the country’s selection pathway for the International Mathematical Olympiad (IMO).

Grants

AWS Cloud Credit for Research [\[link\]](#)

Sep 2026

Research Area: Quantum Computing

Requested Amount: US\$5,000.00 in AWS Braket credits

Status: Submitted September 17, 2026

EAP5054 Entrepreneurial Training for Scientists II (PRPG/USP) *Spring 2025*
[\[link\]](#)

Project Title: StrategiQ — AI-Based SaaS for Quantum Computing Governance Consulting

Role: Founder and Project Lead of a three-member team conducting market research and developing the business plan for StrategiQ developed through the Entrepreneurial Training for Scientists I (DPG5011)

Funding Agency: Office of the Provost for Graduate Studies, University of São Paulo (PRPG/USP)

Total Amount: R\$3,720.00

132734/2025-7 [\[link\]](#)

2025–2027

University of São Paulo (USP)

Program: Master's Scholarship (GM)

Funding Agency: National Council for Scientific and Technological Development (CNPq)

Total Amount: R\$50,400.00

409642/2022-3 [\[link\]](#)

2023–2024

Project Title: Quantum Computing Applied to Molecular Modeling

Program: Technological Development in ICTs (DTC-F)

Funding Agency: National Council for Scientific and Technological Development (CNPq)

Scholarship Process: 301681/2023-6

Total Amount: R\$36,300.00

Software

1. MORAES, N. A.; LUZUM, M. W. *hic-bayes*. [\[Sec. 4.1\]](#)

2026

Python software for Bayesian parameter estimation and Gaussian-process emulation in relativistic heavy-ioncollision studies. Implements PCA-based dimensionality reduction, Gaussian-process emulators, multivariate likelihoods, MCMC sampling, posterior analysis, and multi-fidelity workflows.

2. MORAES, N. A. *Zotero Mirroring Plugin*. [\[link\]](#)

2026

Zotero plugin for mirroring the hierarchical structure of a Zotero library onto the local filesystem.

3. MORAES, N. A. *Git Release and Milestone CLI*. [\[link\]](#)

2026

Command-line tool for automating Git releases, milestones, and software development cycles.

4. MORAES, N. A. *Quantum Battleship*. [\[link\]](#)

2025

Quantum computing game that allows users to play Battleship using real quantum computers.

5. MORAES, N. A. *Molecular Symmetry*. [\[link\]](#)

2025

Web application for parsing molecular geometries from XYZ files and generating molecular symmetry reports.

Leadership Experience

Founder & Team Lead [\[link\]](#)

Fall 2025

University of São Paulo (USP)

DPG5011 – Entrepreneurship

- Founded and led a six-member interdisciplinary team in the development of an innovative startup focused on quantum computing.
- Led the team through the development of the startup concept, including problem definition, business strategy, and the application of quantum technologies to industry-oriented solutions.

Founder & Team Lead [\[link\]](#)

Spring 2025

University of São Paulo (USP)

EAP5054 – Entrepreneurship

- Founded and led a three-member interdisciplinary team in the development of an innovative startup focused on quantum computing.
- Led the team through the development of the startup concept, including problem definition, business strategy, and the application of quantum technologies to industry-oriented solutions.

	Undergraduate Research Mentor <i>Institute of Physics, University of São Paulo (IFUSP)</i> • Mentored a group of five undergraduate students in scientific initiation projects in quantum computing, providing guidance throughout their research activities. • Supervised the development of individual research projects presented by the students at the Undergraduate Research Symposium.	2025
Applications & Proposals	Quantum Computing and Sustainability: Hackathon and Workshop [link] <i>Brazilian Center for Research in Physics (CBPF) – Rio de Janeiro</i> – Application accepted; participation in the Quantum Computing and Sustainability Hackathon confirmed.	Nov–Dec 2026
	Application to MBA Program [link] <i>Harvard Business School (HBS)</i> – Submitted application for admission to the MBA Program.	Jan 2023
Professional Diplomas	Postgraduate Specialization in Financial and Capital Markets [link] <i>Mackenzie Presbyterian University</i> • Quantitative methods, corporate finance, financing, and financial products and markets	2021 – 2022
Professional Experience	Software Engineering Specialist <i>Open Co</i> [link] • Developed and optimized backend services for a digital banking platform, including the first installment payment solution via PIX. • Proposed and prototyped the migration of a credit-analysis workload from Kubernetes (EKS) to AWS Lambda, reducing response time and operational costs by approximately 90%.	2020 – 2022
	Senior Software Engineer <i>TIDEXA</i> [link] • Contributed to the migration of a payroll system from Oracle Forms/PL–SQL to a Java-based platform. • Led technical decisions and development activities within the engineering team.	2019 – 2021
	Competitive Public-Sector Examination Preparation Independent, full-time study • Multidisciplinary preparation for competitive Brazilian public-sector examinations, including Law, Accounting, Economics, Public Finance, Public Administration, and Business Administration.	2016 – 2018
	Mid-Level Software Engineer <i>Itaitec – Grupo ITAUSA</i> [link] • Redesigned a business-critical software application using software design patterns and engineering best practices, improving usability, reliability, and maintainability. The application was deployed across more than 5,000 branches of a major Brazilian bank.	2014 – 2016
	Junior Software Engineer <i>Itaitec – Grupo ITAUSA</i> • Developed and maintained software applications for retail industry.	2012 – 2014
Talks & Presentations	Conference Presentations 1. “The Effect of Correlated Experimental Uncertainties on Bayesian Inference of Quark–Gluon Plasma Properties.” [link] <i>PHENOEXP Workshop</i> Poster presentation, August 2026.	

2. “Study of the Effect of Correlated Experimental Uncertainty on Bayesian Inference of Quark–Gluon Plasma Properties.” [\[link\]](#)
XXXIII Reunião de Trabalho sobre Interações Hadrônicas (RETINHA XXXIII)
Oral presentation, June 2026.
3. “Risk Management and IT Governance in the Adoption of Quantum Computing: A COBIT 5-Based Conceptual Model.” [\[link\]](#)
5th Congress on Risk Management and ERM – FEA–USP
Oral presentation, December 2025.
4. “Exploring the Potential of Quantum Computing in Materials Science.” [\[link\]](#)
Autumn Meeting of the Brazilian Physical Society
Oral presentation (co-author), May 2025.

Industry & Professional Talks

1. “Unveiling Quantum Computing: Practical Applications and Future Possibilities.” [\[link\]](#)
Rio Innovation Week – AI and Meta World
Talk, 2024.
2. “Quantum Computing. Hands On: Grover Search Algorithm.”
Open Co – Tech Day
Talk, 2022. *[Internal material]*
3. “Shift Your Programming Paradigm to a Quantum Mindset.” [\[link\]](#)
The Developer’s Conference (TDC Business)
Talk, 2022.
4. “Fundamentals of Quantum Computing.”
Open Co – Tech Day
Talk, 2021. *[Internal material]*
5. “Technical Software Documentation.”
ITAUTEC
Seminar, 2014. *[Internal material]*

Early Education Highlights

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| <p>School Environmental Research Project – Municipal Waste Management
<i>Colégio Galáxia – Goiânia, Goiás, Brazil</i></p> <ul style="list-style-type: none"> • Conducted a school research project on municipal solid waste management, including an interview with Pedro Wilson Guimarães, then Federal Deputy for the State of Goiás. • Investigated waste-management challenges and technologies, including landfill biogas recovery and international approaches to waste incineration. • Wilson later served as Mayor of Goiânia (2001–2004), where his administration established municipal selective waste collection through legislation enacted in 2002. | 1999 |
| <p>School Environmental Field Visit – Goiânia Municipal Landfill [link]
<i>Escola do Parque [link] – Goiânia, Goiás, Brazil</i></p> <ul style="list-style-type: none"> • Participated in a field visit to the Goiânia Municipal Landfill, observing municipal solid waste disposal operations and environmental-control systems. • Visited the landfill-gas collection and flaring system, where biogas was captured and burned rather than recovered for energy production at the time. • Observed the leachate treatment system used to treat landfill effluent before the treated water was discharged back into the local river system. | 1998 |
| <p>“Aprender a Pensar” – High Abilities/Gifted Education Extension Program
<i>Pontifical Catholic University of Goiás (PUC Goiás) – Goiânia, Goiás, Brazil</i></p> <ul style="list-style-type: none"> • Selected for participation in a university extension program for children identified with high abilities/giftedness through psychometric assessment of non-verbal abstract reasoning, including visual pattern recognition and inductive reasoning tasks. • Participated in enrichment activities designed to develop higher-order reasoning, problem-solving, creativity, and metacognitive skills. | 1996 |

	Logosophical Education – First Grade <i>Colégio Logosófico González Pecotche – Goiânia, Goiás, Brazil</i>	1995
	• Completed first grade at a school whose educational approach was grounded in Logosophy, a system of thought developed by Carlos Bernardo González Pecotche that emphasizes conscious self-knowledge and the development of individual capacities. • The educational experience placed particular emphasis on activities fostering self-awareness, reflection, creativity, and personal expression.	
Certificates	HSK 3 – Chinese Proficiency Exam Scheduled [link] <i>Instituto Confúcio – São Paulo, SP</i>	18 Oct 2026
Extracurricular Activities	1st Place – Brazil Interstate Taekwon-Do ITF Competition [link] <i>Búzios, Rio de Janeiro, Brazil</i>	Aug 2023
	• Two Gold medals: Sparring and Patterns (Tul).	
	1st Place – Children’s Drawing Contest [Art Portfolio] <i>Claro (telecommunications company) – Brazil</i>	Oct 2000
	• First-place award in a Children’s Day drawing competition organized by Claro for employees’ children.	

* Information shown in gray refers to future or pending items.